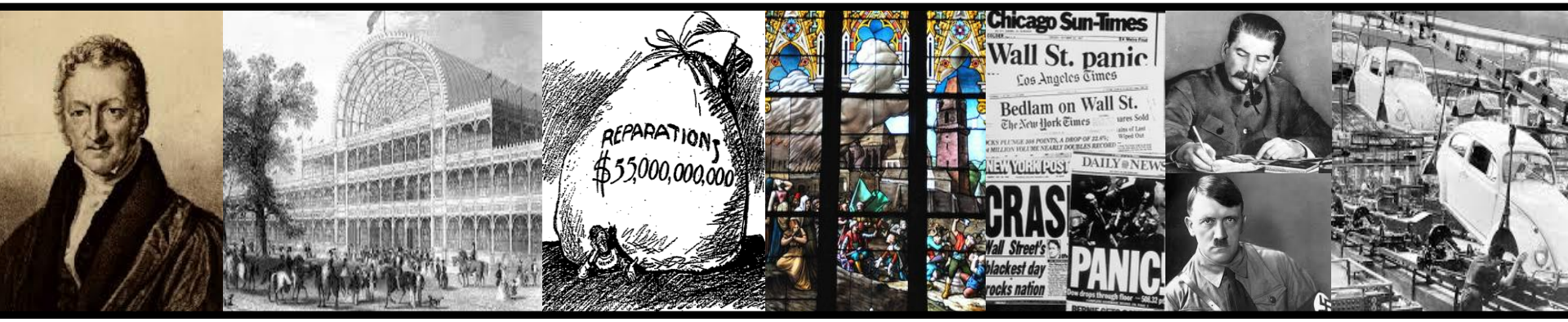


Economic History

BA Lecture — Fall 2026 — by Christian Ochsner (CERGE-EI)

The economics of wars and their consequences

University of St. Gallen, October 16, 2026
(Online session via podcast)

Term schedule Fall 2026

No.	Date	Time	Topic
1	Sept 18	08:15 – 10:00	Overview and Introduction
2	Sept 18	08:15 – 10:00	Pre-industrial Europe: From Malthus to early development
3	Sept 25	08:15 – 10:00	The Industrial Revolution
4 ##	Oct 02	08:15 – 10:00	(Buffer)
5	Oct 09	08:15 – 10:00	Cultural economics
6 ##	Oct 16	08:15 – 10:00	The economics of wars and its consequences
7	Oct 23	08:15 – 10:00	The Great Depression in the US and Europe
	Semester Break (no lectures)		
8	Nov 13	08:15 – 10:00	The economics of totalitarian regimes: Nazis and USSR
9	Nov 20	08:15 – 10:00	Regional economics and ethnic cleansing after WWII
10	Nov 27	08:15 – 10:00	Economic growth after WWII in the East and West
11	Dec 04	08:15 – 10:00	From Bretton Woods to the Euro
12	Dec 11	08:15 – 10:00	Buffer, Repetition, Q&A and exam preparation
13	Dec 18	08:15 – 09:45	Final exam of 90 min (Reading time: 10min; Writing time: 80min)

##: Provision of a Podcast due to absences (tba)

Today

A snapshot on World War I (WWI)

- The beginning of the 2nd "Thirty Years War"
- Stylized facts on the economics of war-capacity
- The consequences of WWI

Reparations, hyperinflation and their ends

- Reparations scheme and hyperinflation in Germany
- The hyperinflations in Germany, Austria, Hungary and Poland
- The end of hyperinflations: Expectations vs. real (monetary) policy shifts

WWI and women empowerment

- Lost men and the intergenerational effects on women: The case of France

(Un)Intended consequences of wars

- Infant industry protection: The case of the Napoleonic blockade

A snapshot on World War I

The economics of the second "Thirty years wars"

- The time from 1914 to 1945 is labelled as "The second thirty years war" (W. Churchill)
  - Meaning WWI and WII including the interwar period (coordination failure and mistrust)
- WWI and WWII were unique in history in terms of the use of economic resources
  - Countries were able to invest over 50% of their GDP for war purposes in WWI and WWII
  - This would have been impossible before 1914 because of low wealth and scarcity of resources
  - After 1945, such GDP ratios were no longer necessary as a result of "cheap" weapons of mass destruction
  - *"In terms of bangs per buck, military technology has never been cheaper"* (Ferguson 2001, p.40)

GDP ratios for war expenses during WWI (in % of GDP):

Year	Australia	Canada	France	Germany	UK	USA
1913	5.5	7.0	10.0	9.8	8.1	1.8
1914	5.7	10.0	22.3	23.9	12.7	1.9
1915	9.6	13.1	46.4	43.8	33.3	1.9
1916	14.0	16.5	47.2	50.3	37.1	1.5
1917	17.2	15.7	49.9	59.0	37.1	3.2
1918	17.2	16.9	53.5	50.1	35.1	16.6

Source: Broadberry und Harrison (2005)

A snapshot on World War I

The economic and political situation before WWI

- Conflicts between the major European powers increased after 1880
  - Divergence in economic development: UK and Germany grew (population, GDP, technology) while France, Austria-Hungary, the Ottoman Empire and Russia fell back (in relative terms)
  - Germany claimed colonies (land, natural resources and markets)
- Increasing tensions within the major powers
  - Nationalist and Social Conflicts in Austria-Hungary, Russia and Germany
  - Balkan conflicts and loyalty to the Alliance then triggered the WWI
  - Clash between the Ottoman Empire, Austria-Hungary and Russia
  - Intervention of Germany, Great Britain, France, (thereafter: USA and Italy)

Europe before WWI:



Source: IEG Mainz

A snapshot on World War I

The economics of wars: Endowment

- Winning a war depends on economic resources and their mobilization for the military
  - Ritschl (2014): For Germany, starting the war in 1914 was the last chance to win it (Thus: No German de-escalation policy after Sarajevo)
- Comparison of the Central Powers with the Allies in 1914, 1916 and 1918
  - Important variables to win a long-lasting war: Population (soldiers), land area (natural resources) and GDP, GDP per capita (technology, possibility of mass production)
  - The Central Powers were able to win the war only through military strategy or surprise (the German Blitzkrieg in WWII), but never as a result of their resources

Resources by the Allies compared to Central Powers in 1914, 1946 and 1918 (Central Powers = 1):

	Population	Total area	Area per capita	GDP	GDP per capita
November 1914					
Total	5.2	11.5	2.2	2.9	0.6
Major powers only	2.2	19.4	8.8	1.8	0.8
November 1916					
Total	5.5	12.1	2.2	3.2	0.6
Major powers only	2.2	19.4	8.8	1.8	0.8
November 1918					
Total	8.1	13.5	1.7	4.6	0.6
Major powers only	1.6	7.5	4.8	2.5	1.6

Source: Broadberry und Harrison (2005)

The consequences of WWI

Political consequences

- Democratization: Extension of the welfare state and female suffrage
  - Welfare state: The mass mobilization of women and men for military service during WWI led to political concessions to the population
  - The welfare state changed the attitude toward migration/migrants (who belongs to us?) which provides a breeding ground for radical ideologies
  - (Promised) Female suffrage from 1918 to 1920: Germany, Czechoslovakia, Hungary, Austria, Ireland, Canada, Poland, Netherlands (later: United Kingdom in 1928 and Switzerland in 1971!)
  - New countries in CEE that also serve a buffer between the major power (Germany vs. USSR)

WWI as a playground for "central planning"

- War mobilization led to the development and application of (central) planning technologies
  - Germany: Walther Rathenau (son of the AEG founder and a leading Siemens manager) designed a model for a planning economy which Lenin (1917) called *"a complete guide to socialism"*
  - The Nazis and the GDR further develop these WWI-Central-Planning instruments

Europe after WWI:



The consequences of WWI

Reparation claims and war debt

- War debt among the Allies
  - The European allies won the war only with the military and monetary(!) help of the USA
  - The UK and France imposed reparation claims to the Central Powers (especially Germany, Austria and Hungary) to repay their debt to the US

Public debt and reparation claims – Germany, UK and France:

	Germany		UK		France	
	In bn Goldmark	in % of GDP in 1913	In bn £	in % of GDP in 1913	In bn FF	in % of GDP in 1913
Public debt in 1913	32.8	63	711	33	33637	67
Public debt in 1920	25.2	48	3160	144	66953	135
A-Reparations	10.0	19				
B-Reparations	42.0	80				
C-Reparations	80.0	152				
Sum of debt	157.2	299				

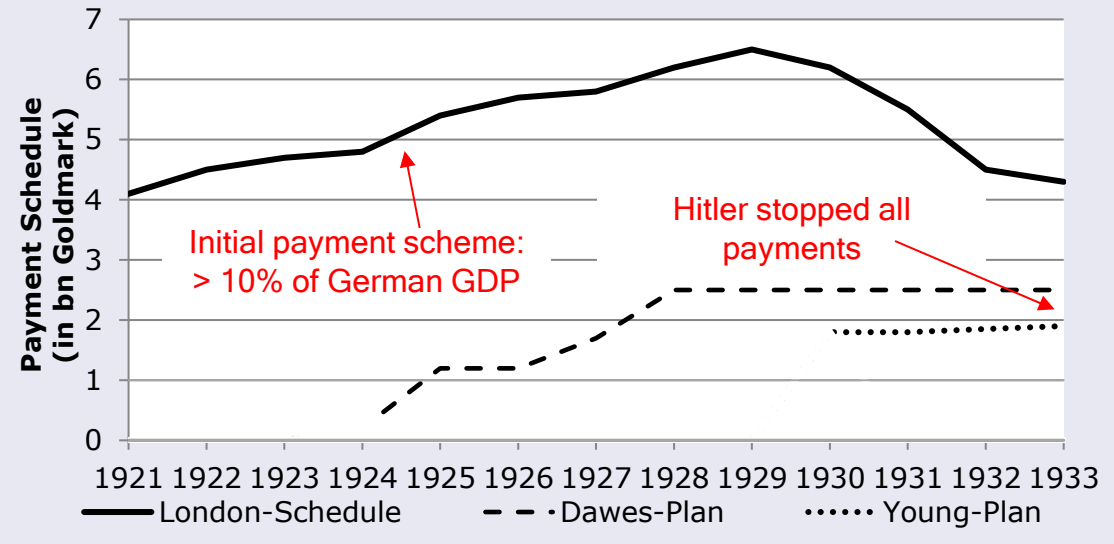
- Reparation payments were resolved in 1920 and fixed in 1921 (to be paid in kind or cash)
  - A-Reparations: To cover direct war costs (amount comparable to historical payments after defeats)
  - B-Reparations: To repay war debts of France and GB to the US (debt was higher than historically due to the "industrialization of war")
  - C-Reparations: Intended as punishment and as propaganda for the "home front"

The consequences of WWI

How can Germany repay its debt and reparation payments (in cash or in kind)?

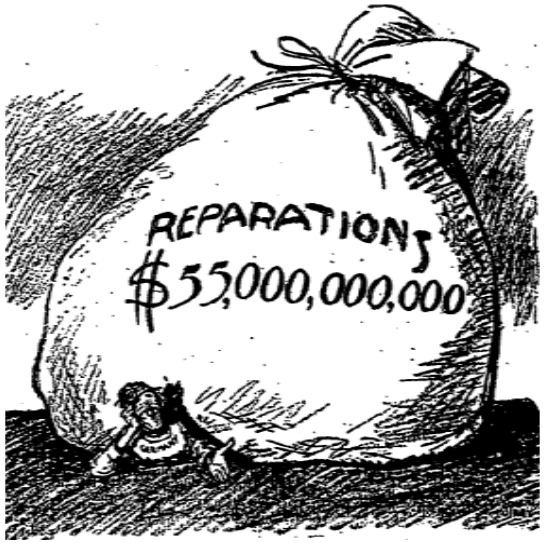
- German citizens will suffer due to higher (consumer) taxes and lower wages in order to produce a current account surplus to pay the Allies
- Terms of trade will be against Germany: Prices of transferred goods will fall relative to the prices of imported goods (a trade surplus is even harder to achieve)
  - J. M. Keynes (1919) concludes in "The Economic Consequences of the Peace": *"Germany can never execute this treaty. Either renegotiation or anger and revenge!"*

Repayment scheme for Germany:



Source: Webb (1986)

Will Germany survive?:



Source: Eichengreen (1992)

Germany: How to pay reparations?

States have three options to meet their payment liabilities (either in domestic or foreign currency):

- (1) Higher (consumer) taxes/VATs and higher costs for public services
  - Higher taxes, especially higher consumer taxes, increases trade surplus: Demand for domestic and foreign goods fall (no go: higher taxes on labor would also increase export prices)
  - Trade/current account surpluses are then used to repay debt ("Typical" IMF recommendation)
  - But who pays: Consumers will be opposed this option
- (2) Issue new debt to pay the reparations (debt rescheduling)
  - But who will lend money to Germany and to what price (interest rate)?
  - Furthermore: Germany needed FDI after WWI: Who will invest in such an environment?
- (3) Print money to pay the reparations/to repay debt
  - Printing money (above the growth rate of the economy) increases the risk of inflation
  - Printing money also feasible if debt is denominated in foreign currency due to seigniorage (Revenue of additional money at given short term exchange rates)
  - The problem of inflation: (i) Affects exchange rates (devaluation of the currency); (ii) Imports become more expensive (unhappy consumers); (iii) Inflation reduces the value of tax revenues from t-1; (iv) Inflation eventually leads to capital flight (and hoarding of value-preserving assets)
  - Thus: Printing money as a last resort !!!

Germany: How to pay reparations?

The hyperinflation in Germany: The process report in a nutshell

- What is hyperinflation?
  - Fast devaluation of the money (equivalent: A rapid increase in the price level)
  - Rule of thumb (Wikipedia): Hyperinflation if prices increase >50% per month (>13000% p.a.)
- The situation in Germany until 1922/23
  - Germany stabilized after WWI in political and economic terms and implemented a successful tax reform: Increasing tax revenues, reducing unemployment, stabilizing prices, economic growth
  - In 1920/21: Reparations-payments were announced that led to political tensions, violence and finally reduced tax revenues
  - Central bank started to finance the budget by printing money in Summer 1921: Start of inflationary pressure and the depreciation of the exchange rate

The German budget in 1922 (in Mio. Goldmark) and the start of the (hyper)inflation:

Quarter	Revenues	Spending	Deficit	Inflation (%)[a]
1922-I	1'205	1'703	499	55.8
1922-II	1'293	1'590	297	29.4
1922-III	888	1'473	585	308.3
1922-IV	646	1'472	826	419.9

a) Quarterly changes in the price level (WPI)

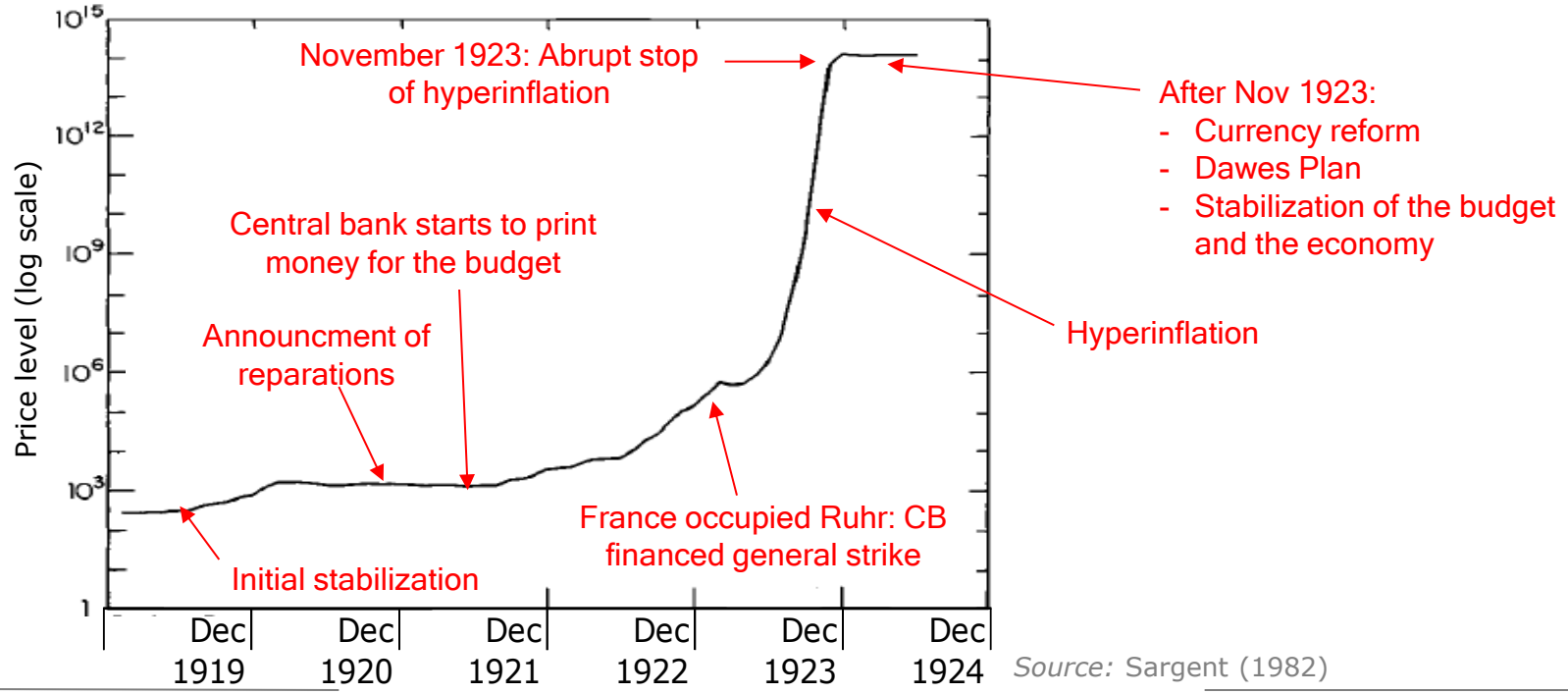
Source: Eichengreen (1992)

Germany: How to pay reparations?

The hyperinflation in Germany: The year 1923

- Germany was not able to pay the reparations anymore: France invaded the Ruhr region (the most industrialized German region) to get the reparations in kind
- The German government co-organized a general strike financed by the Central Bank (workers get paid by the Central Bank): Hyperinflation!

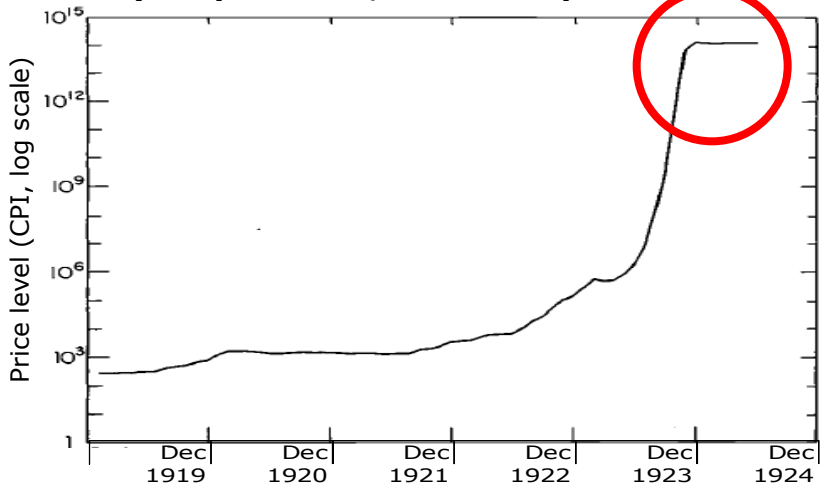
The German (hyper)inflation of 1922/1923:



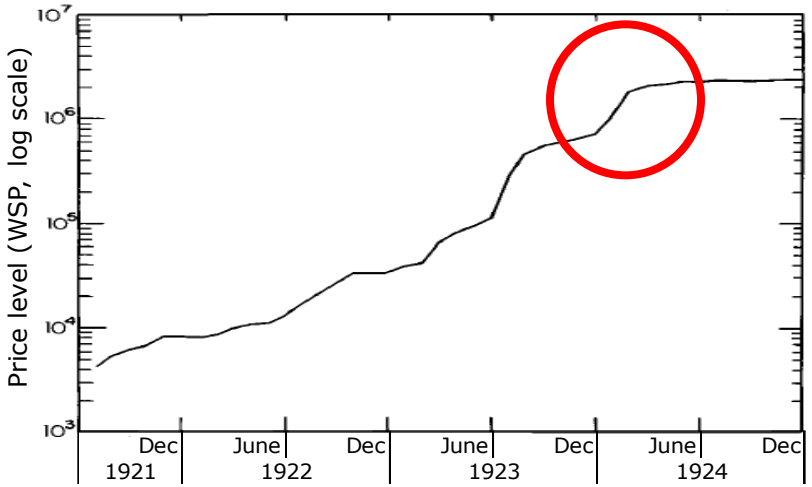
Source: Sargent (1982)

Four hyperinflations and their ends

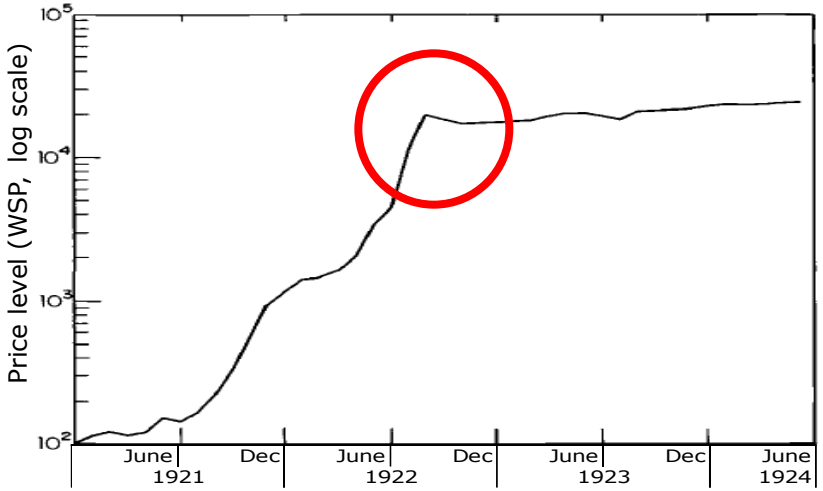
Germany: Reparations, Ruhr-occupation



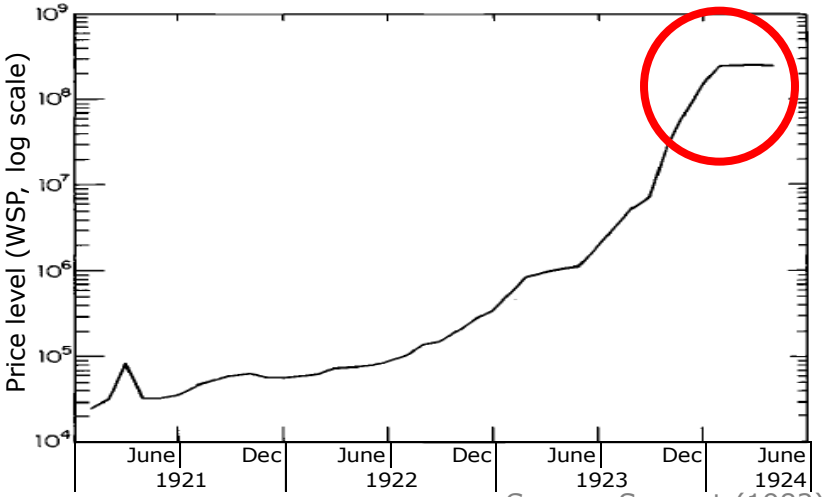
Hungary: Reparations, foreign debt



Austria: Reparations, foreign debt



Poland: War with USSR, 3 different tax systems



Source: Sargent (1982)

Four hyperinflations and their ends

The general pattern of four big hyperinflations

- Inflation was dramatic, but followed a similar pattern:
  - In all four cases: Massive budget deficit, printing money, inflation, depreciation of the currency...
  - But: Hyperinflation stopped suddenly more or less from one day to the another

The drivers of the sudden stop

- Lucas (1976): "Inflation is driven by expected inflation; to end it, change factors that drive expectations"
- Sargent (1982): "Hyperinflation and its end provides strong evidence in favor of rational expectations"

Key issue: How do agents (firms, consumers) form expectations about inflation?

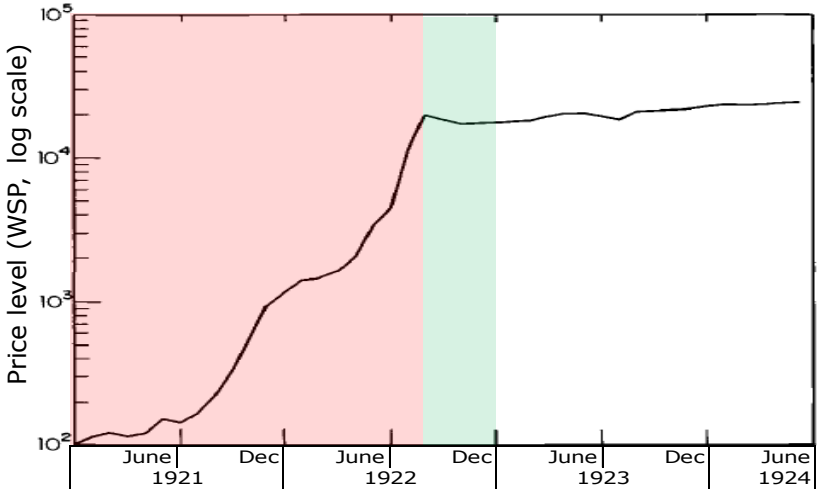
- Adaptive (short-sighted): High persistence in inflation, change in fiscal and monetary policy will be accompanied by only slow reduction in inflation, hence costly
  - US in the mid-1970s: Rising inflation, Keynesian arguments (Philips curve, 1958, 1960) imply high costs to reduce inflation
- Rational: Agents understand mechanism that drives price level; change in that mechanism can have immediate effect on expectations and inflation; policy changes might be cheap

Four hyperinflations and their ends

Austria: Continuing monetary expansion after the sudden stop of hyperinflation

Table A2 **Total Note Circulation of Austrian Crowns (in thousands of crowns)**

1919	January	—
	February	—
	March	4,687,056
	April	5,577,851
	May	5,960,003
	June	7,397,692
	July	8,391,405
	August	9,241,135
	September	9,781,112
	October	10,819,310
	November	11,193,670
	December	12,134,474
1920	January	13,266,878
	February	14,292,809
	March	15,457,749
	April	15,523,832
	May	15,793,805
	June	16,971,344
	July	18,721,495
	August	20,050,281
	September	22,271,686
	October	25,120,385
	November	28,072,331
	December	30,645,658
1921	January	34,525,634
	February	38,352,648
	March	41,067,299
	April	45,036,723
	May	45,583,194
	June	49,685,140
	July	54,107,281
	August	58,533,766
	September	70,170,798
1922	January	227,015,925
	February	259,931,138
	March	304,063,642
	April	346,697,776
	May	397,829,313
	June	549,915,678
	July	786,225,601
	August	1,353,403,632
	September	2,277,677,738
	October	2,970,916,607
	November	3,417,786,498
	December	4,080,177,238
1923	January	4,110,551,163
	February	4,207,991,722
	March	4,459,117,216
	April	4,577,382,333
	May	4,837,042,081
	June	5,432,619,312
	July	5,684,133,721
	August	5,894,786,367
	September	6,225,109,352
	October	6,607,839,105
	November	6,577,616,341
	December	7,125,755,190
1924	January	6,735,109,000
	February	7,364,441,000
	March	7,144,901,000
	April	7,135,471,000
	May	7,552,620,000
	June	7,774,958,000
	July	7,995,647,000
	August	5,894,786,367
	September	7,998,509,000
	October	8,213,003,000
	November	8,072,021,000
	December	8,387,767,000
1925	January	7,902,217,000
	February	7,957,242,000
	March	7,897,792,000
	April	7,976,420,000

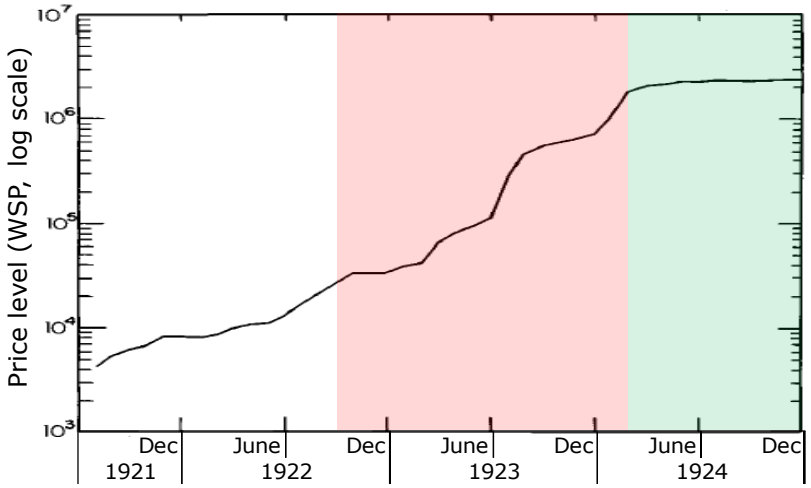


Source: Sargent (1982)

Four hyperinflations and their ends

Hungary: Continuing monetary expansion after the sudden stop of hyperinflation

	Notes in Circulation
October	70,005
November	72,016
December	75,887
1923:	
January	73,717
February	75,135
March	82,205
April	100,101
May	119,285
June	154,996
July	226,285
August	399,487
September	588,810
October	744,926
November	853,989
December	931,337
1924:	
January	1,084,677
February	1,278,437
March	1,606,875
April	2,098,091
May	2,486,257
June[1]	2,893,719
July	3,277,943
August	3,659,757
September	4,115,925
October	4,635,090
November	4,442,644
December	4,513,990

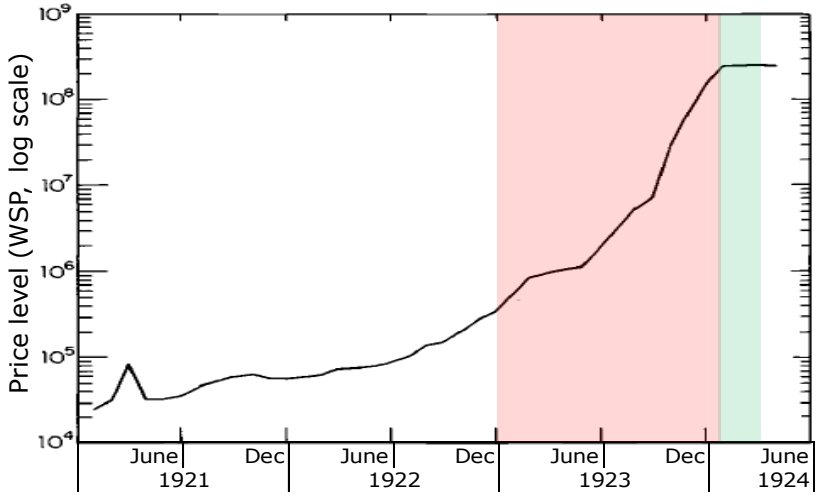


Source: Sargent (1982)

Four hyperinflations and their ends

Poland: Continuing monetary expansion after the sudden stop of hyperinflation

Month	Note Circulation
1923:	
January	909,160.3
February	1,177,300.8
March	1,841,205.6
April	2,332,396.8
May	2,733,794.1
June	3,566,649.1
July	4,478,709.0
August	6,871,776.5
September	11,197,737.8
October	23,080,402.2
November	53,217,494.6
December	125,371,955.3
1924:	
January	313,659,830.0
February	528,913,418.7
March	596,244,205.6
April	570,697,550.5

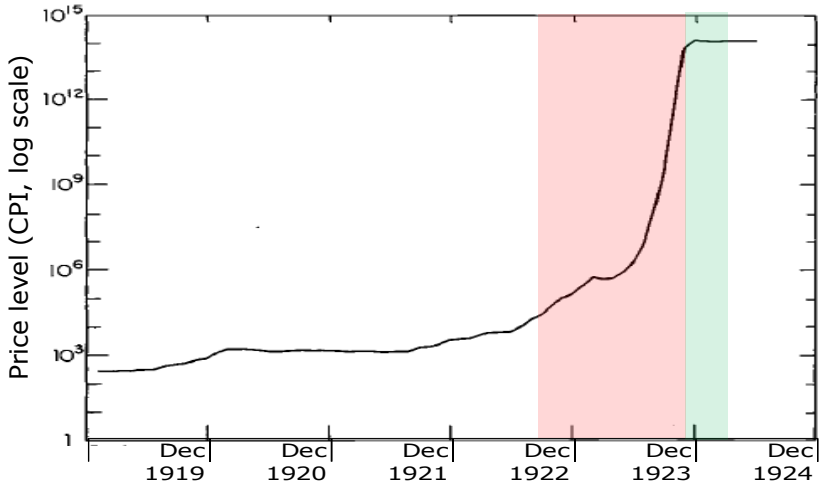


Source: Sargent (1982)

Four hyperinflations and their ends

Germany: Continuing monetary expansion after the sudden stop of hyperinflation

Month	Note Circulation
October	469,456,818
November	754,086,109
December	1,280,094,831
1923:	
January	1,984,496,369
February	3,512,787,777
March	5,517,919,651
April	6,545,984,355
May	8,563,749,470
June	$17,291^2$
July	$43,595^2$
August	$663,200^2$
Setember	$28,228,815^2$
October	$2,496,822,909^2$
15 November	$92,844,720,743^2$
30 November	$400,267,640,302^2$
December	$496,507,424,772^2$



Source: Sargent (1982)

Four hyperinflations and their ends

Four hyperinflations ended due to regime change

- Germany: Currency reforms and foundation of the Rentenbank with binding limits on government borrowing and issue of money (backed by land and other assets)
  - This was accompanied by the new Dawes-Plan on reparations that lowered the immediate burden as well as the total burden of reparations
  - Intervention by the League of Nations
  - Large international loan and FDI
- Austria, Hungary, Poland stabilized currencies with support from the League of Nations, international loans and monetary reform (partial independence of Central banks)
  - For Poland: Peace settlement with USSR was crucial
- All four countries entered the gold-exchange standard with fixed exchange rates
  - This increases the credibility of the central banks/of monetary policy and made foreign direct investment (FDI) more likely
- Is there any rationale behind the regime change?
  - Kydland & Prescott (1977): "The time inconsistency problem"

Fragile stabilization in 1924

1924 looked better than it was

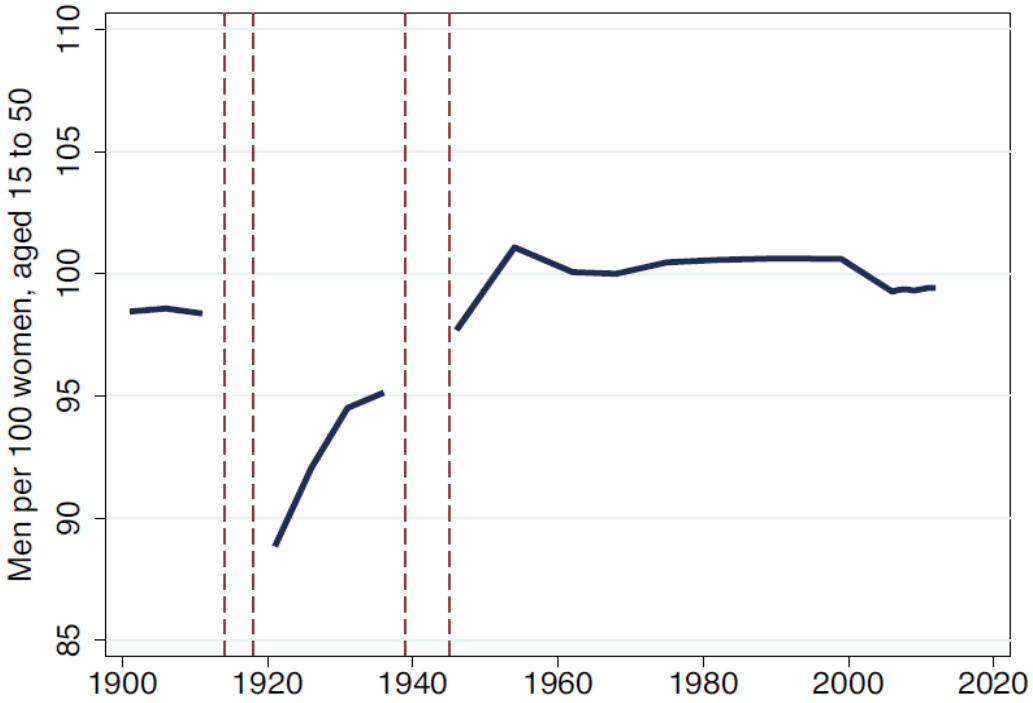
- Everything looked well in 1924: Political stabilization, price stabilization, economic upturn, better but still imperfect conditions for Germany
- But: Key problems were postponed, not really resolved:
  - Dawes-Plan as interim solution to calm the waters and find a new reparations settlement; linked to open issue of war-debt
  - Stabilization triggered wave of short-term (hot) capital inflows into Central Europe and Germany
- Gold-exchange standard inherently problematic, because:
  - Parities did not match market rates (Pound overvalued, France still struggling 1924-26, enters GS at undervalued rate)
  - Backing of currencies by foreign exchange as a new transmission channel
  - Lack of Central Bank cooperation (gold hoarding), esp. in countries with inflationary experience
  - New political role for the working class: GS gives less space for country specific needs (as now in the Euro area)

WWI and women empowerment

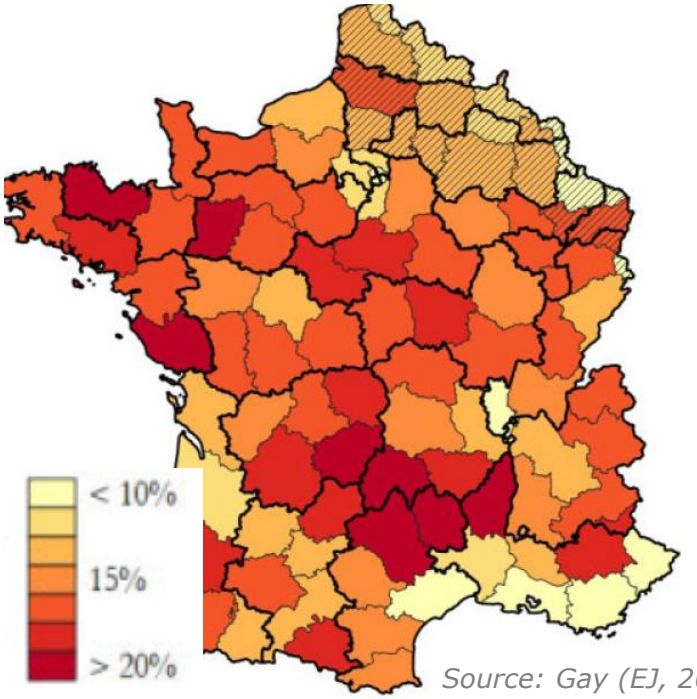
Women increased their social standing with war-affectedness

- Right to vote: Mainly and firstly in countries heavily affected by the war
- Economic standing: Lost men gave space for increased women labor force participation
- The example of France (see Gay, EJ 2023):

Gender ratios:



Death rates (county level):



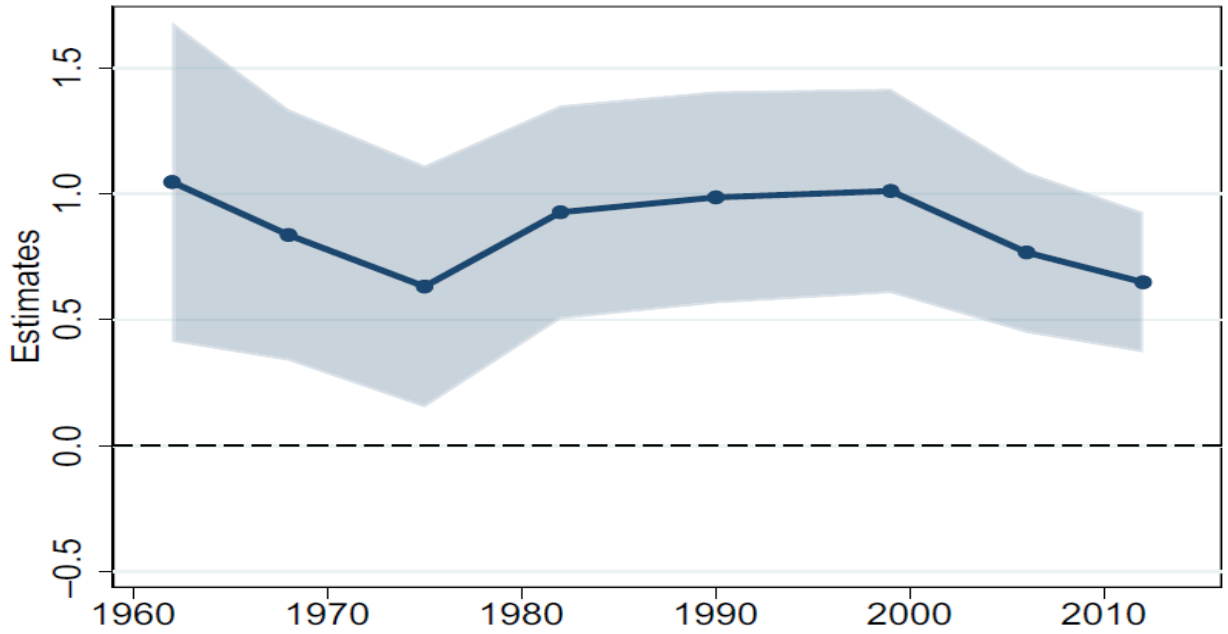
Source: Gay (EJ, 2023)

WWI and women empowerment

Women empowerment and long-run outcomes of labor force participation

- Counties with higher casualty rates bring immediately more women into the labor force
  - Only look at "migrating women": Residing and working in region *i*, but born in region *j* to exclude effects of local labor market structure
  - This effect persists until today and are transmitted (i) from mothers to daughters, (2) from mothers-in-law to daughters-in-law via their sons, and (iii) through local social interactions.

Main results: "Elasticities of casualty rate on women at work (aged 30-49):



Outcome: employed.Treatment: military death rate (county of residence).
Sample: married women aged 30–49, censuses 1962–2012.

Source: Gay (EJ, 2023)

(Un)intended consequences of wars

Can wars be good for a country?

- Conflicts often affect the economy
  - International wars: Destruction, re-allocation of resources, shock in import-export linkages, etc.
  - Sanctions against regimes: Specific persons or industries are sanctioned
- Potential positive effects of wars and sanctions, if any:
  - Large investments into R&D: i.e., Radar, nuclear power, …
  - Certain industries might benefit: i.e., Russian cheese factories started to produce French cheese
- The paper *"Temporary Protection and Technology Adoption: Evidence from the Napoleonic Blockade"* by Juhász (AER 2018) asks whether the Napoleonic blockade spurred industrialization in French regions that were heavily affected by the embargo
- Main findings: French regions that were heavily exposed to the trade embargo build up their own textile industries during the blockade
  - These regions remain more „industrialized" for another 50 years
  - Strong evidence for „infant industry protection" to spur industrialization

(Un)intended consequences of wars

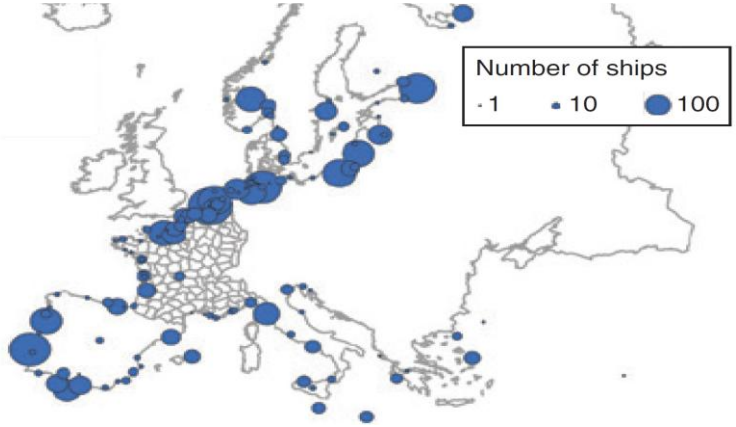
Historical background and the setting

- The Napoleonic wars: Napoleon conquered Continental Europe
  - Britain remained unaffected but aimed at hurting France with a trade embargo
  - Before the embargo: France imported mainly textiles from Britain due to its weak industrial base
- The estimation of the "causal effect" of infant industry protection is almost impossible
  - Protected industries are not randomly chosen by the government (i.e., sorting)
  - In case of success: We do not know whether a protected industry would have become successful anyway (infant industries policies are targeting the entire country – thus: no counterfactual)
- The embargo affected French regions differently
  - North/northwest: Trade distance to Britain increased heavily
  - Southern and central parts of France were relatively less affected by the embargo
- How to measure industrialization due to the embargo
  - Juhász (AER 2018) uses the number of mechanized cotton spindles per thousand inhabitants
  - Mechanized cotton spindles a good measure of physical capital in textile industries
  - Note: Spindles were heavily used in British textile industries at that time, but less so in France
  - Compare the industrialization among heavily and less heavily affected French region before and after/during the embargo in a Difference-in-differences (DiD) framework

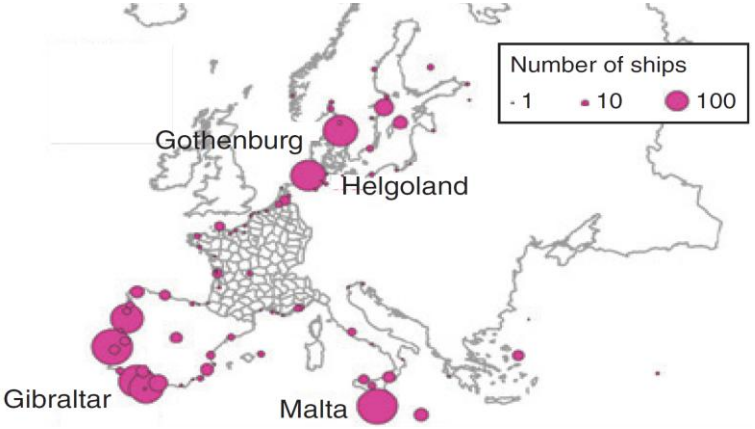
(Un)intended consequences of wars

The embargo and the geography of the French textile industry (1803 and 1812)

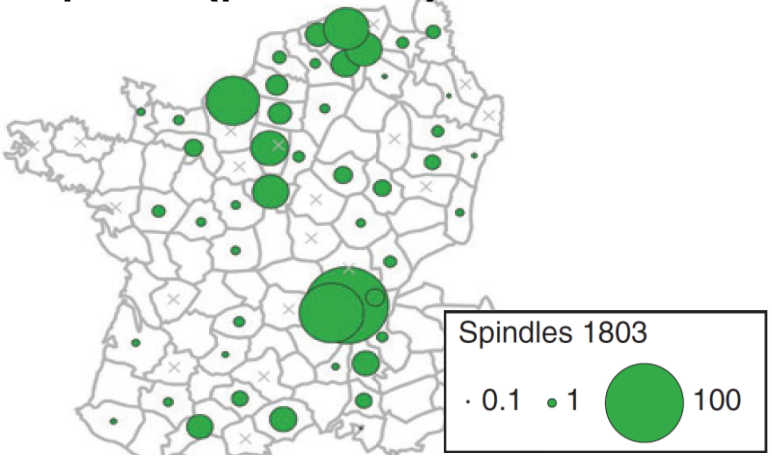
Trade with Britain before the blockade:



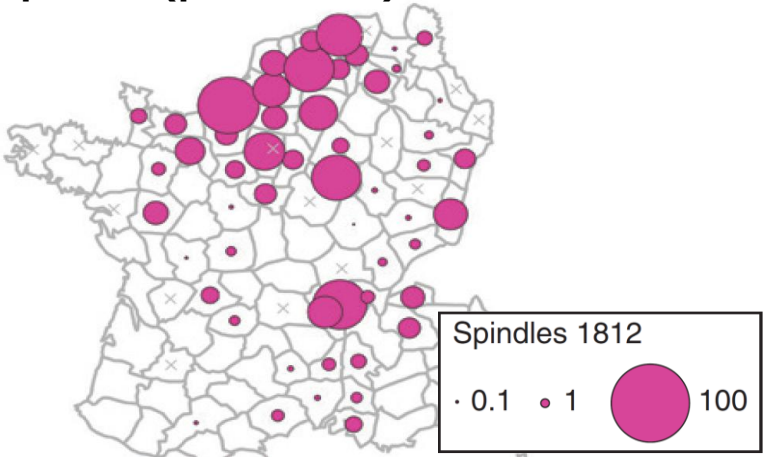
Trade with Britain during the blockade:



Spindles (per 1000 c.) in 1803:



Spindles (per 1000 c.) in 1812:

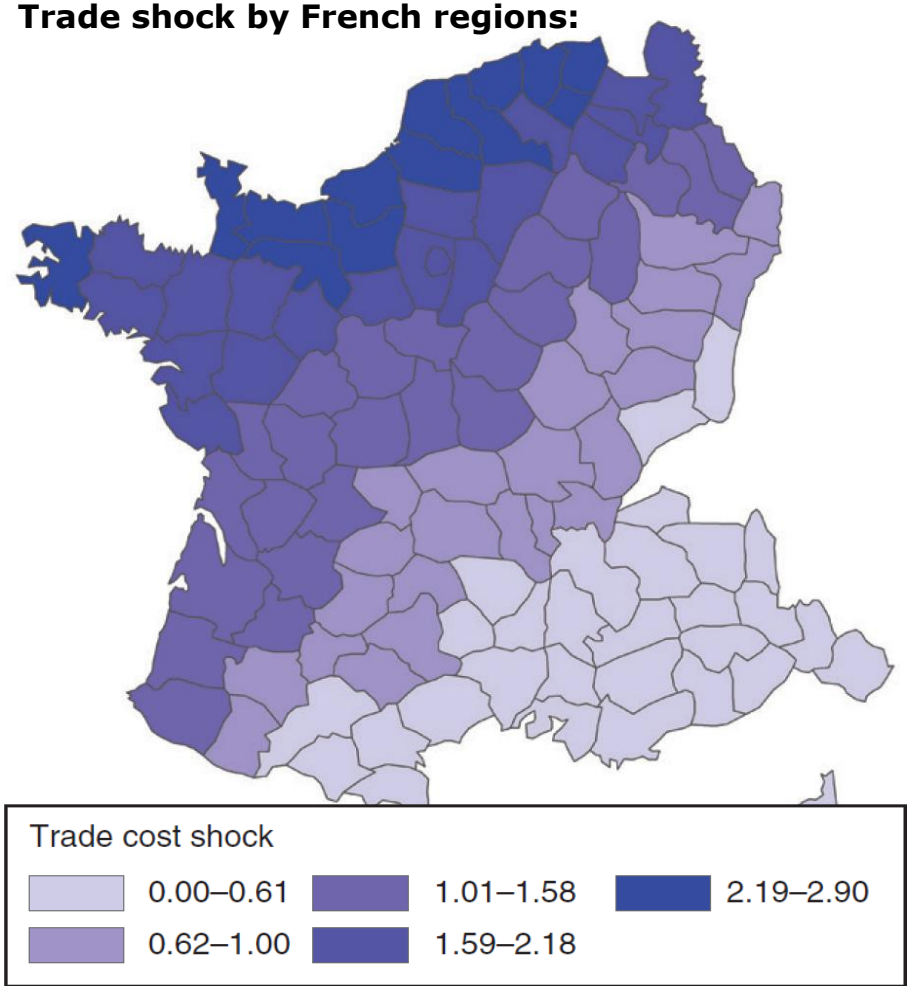


Source: Juhász (2018)

(Un)intended consequences of wars

„Exogenous“ variation in trade costs and results

Trade shock by French regions:



DiD estimates (no. of spindles per 1.000 inhabitants):

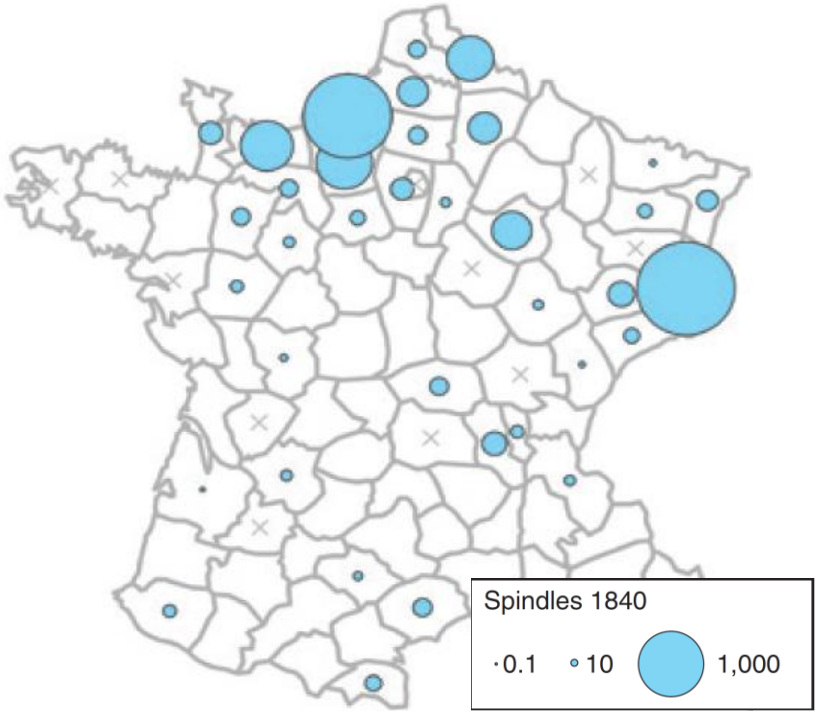
Effective distance	33.47	38.82
	0.47	*0.48*
	(9.80)	(13.23)
	{10.00}	{13.46}
Streams × 1812		−1.16
		(2.17)
Coal × 1812		4.11
		(7.47)
Market potential × 1812		30.19
		(30.19)
Knowledge access × 1812		34.90
		(21.79)
Literacy × 1812		27.79
		(18.86)
Time fixed effects	Yes	Yes
Department fixed effects	Yes	Yes
Observations	176	126
Adjusted R^2	0.34	0.45
Num. clusters (dept)	88	63
Num. clusters (gen)	40	30

Source: Juhász (2018)

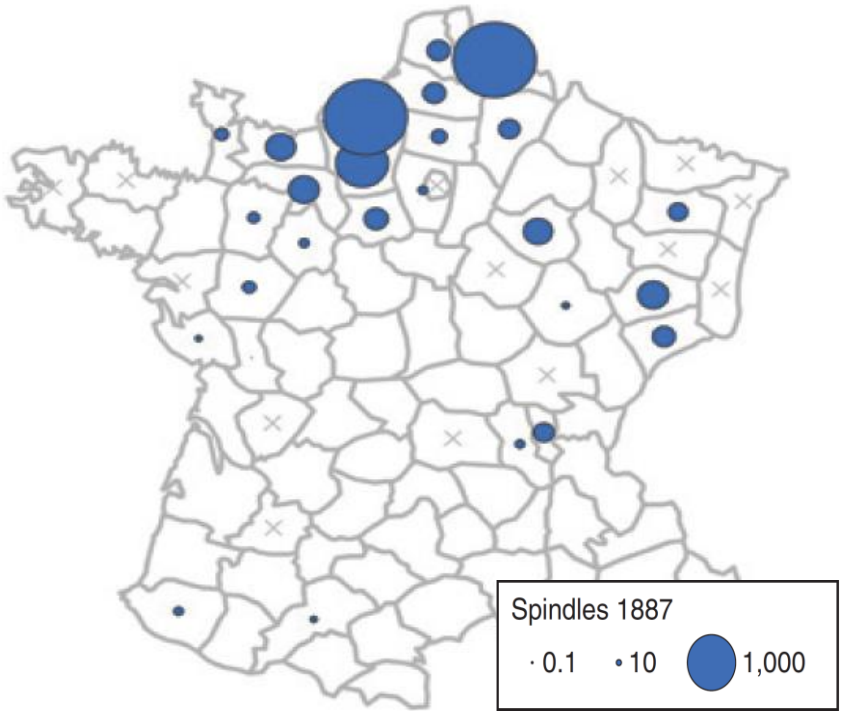
(Un)intended consequences of wars

Long-run effects of industrialization due to the embargo

Spindles (per 1000 c.) in 1840:



Spindles (per 1000 c.) in 1887:



Source: Juhász (2018)

- The North remain the textile hotspot in France for almost a century
  - Juhász (AER 2018) also provides evidence that productivity (value added per capita) was positively shaped for several decades